

eReserve: An Online Facility Reservation System of One Philippine Higher Education Institution

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Asia Pacific Journal of
Management and
Sustainable Development

Vol. 13 No. 1, pp. 28-33

March 2024 Part 2

P-ISSN 2782-9332

E-ISSN 3028-2632

ISSUE DOI:

10.70979/IBHI3176

Abstract – *The objective of this project was to create a digital platform that would optimize the manual facility reservation process at Lyceum of the Philippines University – Batangas (LPU-B). The existing system had two areas from improvement: varying facility availability and overlapping reservations, which impacted productivity. The system, developed in PHP and MySQL, therefore eliminates scheduling problems through reservation booking, instant availability check, and confirmations. Testing proved that the system did not only enhance the user experience but also reduced the errors made with increased accuracy. Furthermore, it also brought about administrative benefits such as notifications and information on facility utilization. This system also becomes a model for other organizations that would improve their resource management and is aligned with the trend of higher education increasingly implementing digital methods.*

Keywords – *administrative efficiency, digitalization, Facility reservation system, online platform, resource management*

Cite this article as: Vega, M. M. & Generoso II, M. J. I. A. (2025). eReserve: An Online Facility Reservation System of One Philippine Higher Education Institution. *Asia Pacific Journal of Management and Sustainable Development*, 13 (1), pp. 28-33. <https://doi.org/10.70979/KZDV1086>

INTRODUCTION

Among the dynamic and very active members of a university, the staff and faculty usually find it difficult to have an entry point to access various facilities like halls and gyms. This is exactly the situation in Lyceum of the Philippines University – Batangas (LPU-B), wherein facility reservation demands have increased and needs a more reliable and effective system. The university uses facilities for everything from academic events and sports practices to meetings and social gatherings. Ensuring that all the facilities are being managed effectively is crucial for the smooth operation of university life. Given that the current manual reservation system has been

working, it is not well-positioned or able to keep pace with growing volumes and the associated complexities of reservations. This involves visiting or calling the administration block to check on availability and, upon satisfaction, reservation appointments; hence, making the process tedious and time-consuming.

This manual reservation system has been used and followed by LPU-B for many years. Though functional, it is still considered tedious and laborious with errors. These errors frequently include conflicting reservations, wherein multiple parties reserve the facility at the same time, and misinformation about facility availability, wherein outdated records lead to false information that causes misunderstandings and inconveniences. These lapses in its manual system for reservation have become more evident as the university tries to move forward and develop. This can be directly impacting to the university community, therefore, wearing them out in many ways to a point that makes things very frustrating and reducing productivity. The kind of inefficiency in an academic setup, where time is critical and resources are very costly, is very detrimental. The current reservation process has the potential to drastically reduce output and satisfaction, which may have an impact on how faculty and staff see their time at the university.

While there are numerous studies and existing systems focused on online facility reservations [1] - [5], there is a lack of tailored solutions specifically designed for the unique needs and complexities of university environments like LPU-B. There is a lack of tailored solutions specifically designed for the unique needs and complexities of university environments like LPU-B. Current systems often do not address the specific challenges faced by university staff and faculty in managing diverse facility types and schedules. For instance, staff often struggle with coordinating the availability of the facilities leading to scheduling conflicts and underutilization of resources. Additionally, the manual handling of special requests, such as equipment setup or requests arrangements, complicates the reservation process and increases administrative overhead. These inefficiencies can result in lost

productivity and dissatisfaction among users. This research aims to fill this gap by developing a customized online reservation system that caters specifically to the requirements of LPU-B, providing a scalable and efficient solution that can be adapted to similar educational institutions.

The purpose of this capstone project was to create an online platform that streamlined the facility reservation process at LPU-B to address these challenges. The goal of this project was to develop a system that offered real-time facility availability, removed conflicts with reservations, and enhanced user experience. Using PHP as the programming language, the researcher planned to create a system that incorporated a calendar for accurate scheduling and reservation administration. Users made reservations, checked the status of the facilities, and received confirmations using this online platform, all from the comfort of their personal technology. The end-user was the focus of the system's design, making it simple to use and navigate for all university personnel, including instructors, staff, and students.

The significance of this study lies in its potential to transform the way facility reservations are handled at LPU-B. By moving to an online reservation system, the university will not only reduce the administrative burden on its staff but also enhance the overall efficiency and productivity of its operations. This digital transition aligns with broader trends in higher education towards digitalization, emphasizing the crucial role of technology in improving administrative processes and enhancing the academic experience. Furthermore, the implementation of this system reflects LPU-B's commitment to innovation and excellence in resource management, setting a benchmark for other educational institutions.

OBJECTIVES OF THE STUDY

The system aimed to develop an online facility reservation system. Specifically, the study aimed to:

1. Design and develop an online facility reservation system to simplify and enhance the efficiency of the booking process.
2. Provide real-time facility monitoring system.
3. Provide and generate facility utilization report.

MATERIALS AND METHODS

The development of the eReserve system followed the phases of the Iterative Model in the Software Development Life Cycle (SDLC). Each cycle involved progressive refinements—from planning and requirements analysis to design, testing, and

deployment. As part of the system design process, the interactions between the key users (i.e., requesters, department heads, and administrators) were mapped out using a data flow diagram to clarify how reservations move through the system.

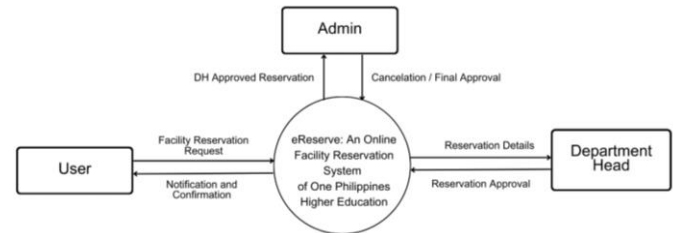


Fig. 1. Level 0 Data Flow Diagram

This diagram provides a high-level overview of the eReserve system's core processes. Users initiate facility reservation requests, which are forwarded to the Department Head for review. Upon approval, the Admin handles the final approval or cancellation. The system also manages report generation, sends notifications to users, and ensures real-time updates on reservation status. This figure illustrates the central interactions between users, department heads, and administrators within the system environment.

Research Design

This project utilized a software development research design, guided by the Iterative Model of the Software Development Life Cycle (SDLC). This approach aligns with recent software development practices in academic institutions [2], [3], [6]. The iterative approach allowed for continuous refinement of the eReserve system through repeated cycles of design, testing, and implementation. This model was chosen to effectively develop a functional and user-centered web-based facility reservation system for Lyceum of the Philippines University – Batangas (LPU-B).

Sampling Technique

The study employed purposive sampling to identify participants who were directly involved in the facility reservation process at LPU-B. These included representatives from the Physical Facilities Management Office (PFMO), faculty members, and administrative staff. Their roles and familiarity with the manual reservation system made them ideal contributors to the requirements gathering, system testing, and validation processes.

System Development Procedure

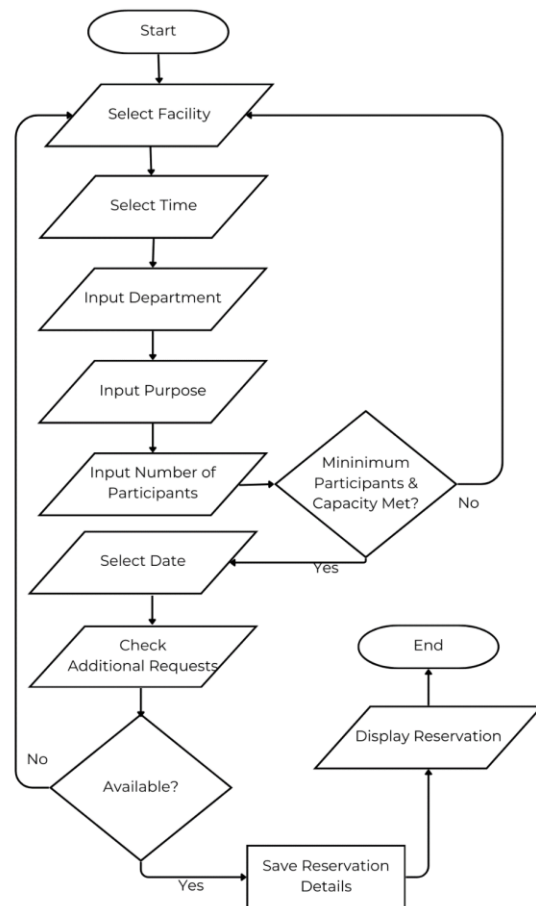
The development of the eReserve system followed the phases of the Iterative Model in the Software Development Life Cycle (SDLC). The process began with planning sessions involving stakeholders to identify inefficiencies in the existing manual reservation system and outline the objectives of the new platform. During the requirements analysis phase, both functional and non-functional needs were gathered through consultations, which helped define the scope and constraints of the project. In the design phase, the logical structure of the system was documented using data flow diagrams (DFDs) that visualized how user actions and system processes were interconnected—particularly how reservation requests moved through approval channels. These diagrams served as blueprints for both developers and stakeholders to agree on system behavior prior to actual coding. Mock-ups were also created using design tools to represent the system's look and feel.

Implementation followed with the use of PHP and MySQL for back-end operations and data management, ensuring dynamic interaction between users and stored reservation data. The interface was developed using HTML, CSS, JavaScript, and Bootstrap, resulting in a responsive and accessible system across desktop and mobile devices. A central component of the implementation is the reservation module, where the system's core function comes to life. As illustrated in Figure 2, the user reservation interface is designed for simplicity and efficiency—allowing users to choose a facility, define time and date preferences, and submit requests in a few streamlined steps. The layout is intentionally intuitive, minimizing training requirements and reducing errors, especially for university personnel who frequently access the system. Visual clarity, logical grouping of inputs, and real-time validation features support a fluid user experience that reflects modern usability standards.

This user-centered design approach ensured that system interactions aligned with the habits and expectations of its intended users. The reservation module is not just functional but also adaptable, supporting various facilities and schedules without overwhelming the user. Testing of this component involved multiple iterations of black-box evaluations and device compatibility checks, with refinements made based on stakeholder feedback. After deployment on the university's official portal (myportal.lpubatangas.edu.ph), the module continued to receive incremental updates to optimize speed,

reliability, and usability based on post-launch observations.

Fig. 2. User Reservation Flowchart



Instrument Validation, Testing, and Data Analysis

The validation of the eReserve system was primarily conducted through qualitative user feedback gathered during alpha testing and trial deployment phases. This approach, consistent with recent facility management research that emphasizes iterative feedback and usability testing, helped ensure the system's reliability, functionality, and user satisfaction [1], [5]. Key stakeholders, including representatives from the Physical Facilities Management Office (PFMO) and selected faculty members, systematically evaluated various aspects of the system such as ease of use, responsiveness, and overall operational stability. Their structured and detailed feedback provided valuable insights into areas needing improvement and guided essential modifications that enhanced the system's performance and user experience. Alongside qualitative feedback, quantitative data were collected from system-generated reports detailing user interactions, reservation patterns, and

overall system usage. This dual method of data collection aligns with well-established practices in software project evaluation within academic and organizational contexts [3], [4]. The qualitative data were subjected to thematic analysis to uncover usability challenges and to inform potential feature enhancements, while the quantitative metrics were analyzed using descriptive statistics to measure the efficiency gains, error reduction, and effectiveness of the eReserve system in managing facility reservations. Together, these data collection and analysis techniques provided a comprehensive understanding of the system's impact and performance, ensuring that the final product met the practical needs of its users while maintaining high standards of quality and reliability.

RESULTS AND DISCUSSION

The development and implementation of the eReserve system for Lyceum of the Philippines University – Batangas (LPU-B) demonstrated significant improvements in the facility reservation process compared to the previous manual method. Results were obtained through user validation, testing sessions, and actual system deployment, which highlighted usability, functionality, and efficiency enhancements.

System testing revealed that the features performed as intended across multiple devices and platforms. Stakeholders reported that the reservation procedure had become more convenient, secure, and efficient. The PFMO confirmed that the system helped reduce double bookings [7], [8] and simplified the tracking and approval of facility requests. Additionally, users noted improved accessibility due to the web-based design, which allowed them to submit and monitor reservations remotely.

Feedback gathered during validation testing also indicated that the layout and interface were user-friendly. Faculty and administrative staff appreciated the dashboard design and notification system, which provided real-time status updates on reservation requests. These results align with previous findings by Chiew and Zainal [1], who highlighted the advantages of web-based platforms in facilitating laboratory reservations and access monitoring.

From a technical standpoint, the PHP-based backend and MySQL database effectively handled the data storage and retrieval requirements of the system. This architecture, combined with front-end responsiveness

supported by Bootstrap, enabled users to interact with the system without noticeable lag or errors. The results support the claim of Lalitha et al. [2] that implementing online booking systems improves operational efficiency and user satisfaction in academic institutions.

Table 1. Comparison Between Manual and eReserve System

Criteria	Manual Process	eReserve System
Booking Method	Walk-in / Phone call	Online via myportal.lpubatangas.edu.ph
Confirmation Time	1–3 days	Instant (Email + System Notification)
Risk of Double Booking	High	Eliminated via real-time availability check
Record Keeping	Manual logs and papers	Automated logs with database reports
User Accessibility	Office hours only	24/7 online access
Special Requests (e.g. equipment)	Manually processed	Logged with each reservation

This table presents a side-by-side comparison of the manual facility reservation process previously used at LPU-B and the newly implemented eReserve system. Key improvements include real-time booking, reduced risk of conflicts, and enhanced accessibility for users through a centralized online platform.

Table 2. System features of eReserve and their Core Functions

Feature	Description
Reservation module	Allows users to request and cancel bookings with preferred schedules
Approval workflow	Multi-layered approval process (Department Head → Admin)
Notifications	Email alerts and status updates to inform users of request status
Admin dashboard	Provides oversight, access control, and real-time system activity monitoring
Facility utilization report	Generates downloadable records of reservation history and usage data

This table outlines the core features integrated into the eReserve system and describes their functions. Each component—from reservation handling to automated reports—contributes to streamlining facility management and reducing the administrative burden at the university. The report generation module (Figure 3) enables administrators to retrieve detailed and utilization data for selected facilities, while the graph report feature provides a visual summary of reservation trends. These functionalities support efficient monitoring and strategic planning in university facility management.

Fig. 3. Report and Graph Generation

This figure displays the report and graph generation modules of the eReserve system. Administrators can select multiple facilities, set a specific date range, and generate detailed or utilization reports. The graph generation section provides a visual overview of reservation trends over time. These tools allow for comprehensive, data-driven decision-making and promote transparency in facility usage at Lyceum of the Philippines University – Batangas.

CONCLUSION AND RECOMMENDATION

The eReserve system successfully addressed the inefficiencies of the manual facility reservation process at Lyceum of the Philippines University – Batangas. By digitizing reservation workflows, the system enhanced

accessibility, reduced booking errors, and improved administrative tracking. User feedback confirmed that the platform was intuitive, secure, and aligned with the needs of faculty and administrative personnel.

Given its positive impact, it is recommended that LPU-B fully adopt the eReserve system for all facility reservations. Future enhancements may include mobile optimization, integration with campus calendars, and expanded analytics to support administrative decision-making. Continuous feedback from users should also guide ongoing system improvements [9]. The success factors and challenges observed in this study align with those reported in similar IT projects [10].

AI TOOL DISCLOSURE.

During the preparation of this work, the author used ChatGPT by OpenAI primarily to check grammar and improve clarity of expression. The content was developed independently, and the author carefully reviewed and edited all AI-assisted text to ensure accuracy and originality. The author assumes full responsibility for the final version of the publication. This disclosure is provided in the interest of academic transparency and integrity.

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